

Binomial Edge Ideals

Final Presentation

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Polymath Jr. Conference

August 8, 2026

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Background

Project Overview

Our project uses concepts from **algebra**, **combinatorics**, and **graph theory**.

We use **binomial edge ideals** to detect underlying structures in graphs. We analyze various examples of graphs, to extend work done for complete graphs.

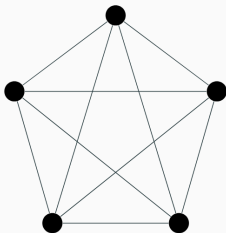


Figure 1: Complete Graph K_5

Monomials and Polynomials

A **monomial** in the set of variables $x_1, \dots, x_n$ is of the form $x_1^{a_1} \cdots x_n^{a_n}$ with $a_i \in \mathbb{N}$. Using shorthand notation with $\mathbf{x} = (x_1, \dots, x_n)$ and $\mathbf{a} = (a_1, \dots, a_n)$

$$\mathbf{x}^{\mathbf{a}} = x_1^{a_1} x_2^{a_2} \cdots x_n^{a_n}$$

The **degree** of a monomial is defined to be the sum of the elements of $\mathbf{a}$.

Example

$x_1^2 x_2 x_3$ is a monomial with $\mathbf{x} = (x_1, x_2, x_3)$, $\mathbf{a} = (2, 1, 1)$ with degree $2 + 1 + 1 = 4$.

Polynomials

A **polynomial** $f(\mathbf{x})$ is a $\mathbb{K}$ -linear combination of monomials. A **homogeneous** polynomial has monomials which are of the same degree.

Example

The polynomial $f(x) = 2x_1x_2^2 + x_1 - 1$ has $\mathbf{x} = (x_1, x_2)$ with $\mathbf{a}$ values

$$\mathbf{a}_1 = (1, 2), \quad \mathbf{a}_2 = (1, 0), \quad \mathbf{a}_3 = (0, 0)$$

f is not homogeneous since the monomials have degrees 3, 1 and 0.

Polynomial Rings

For a field $\mathbb{K}$, the following set

$$\mathbb{K}[x_1, \dots, x_n] = \{p(\mathbf{x}) : p(\mathbf{x}) \text{ a polynomial in variables } x_1, \dots, x_n \\ \text{with coefficients in } \mathbb{K}\}$$

with operations of addition and multiplication is called the **polynomial ring** on n variables with coefficients in $\mathbb{K}$.

Polynomial rings are commutative; for $p_1, p_2 \in \mathbb{K}[x_1, \dots, x_n]$

$$p_1 + p_2 = p_2 + p_1, \quad p_1 p_2 = p_2 p_1$$

A (nonempty) $I \subseteq R$ is called an **ideal** if it satisfies closure under addition and multiplication, that is

- ① if $f, g \in I$, then $f + g \in I$
- ② if $r \in R$ and $f \in I$ then $rf \in I$.

Ideals, Generated

The **ideal generated by** $g_1, \dots, g_s$ in R is the set of finite R -linear combinations of $g_1, \dots, g_s$, denoted

$$(g_1, \dots, g_s) = \{r_1 g_1 + r_2 g_2 + \cdots + r_s g_s \mid r_i \in R\}.$$

Example

In $\mathbb{K}[x_1, x_2]$

$$x_1^3 + x_1 x_2^3 \in (x_1^2, x_2^2) \quad \text{since } x_1^3 + x_1 x_2^3 = x_1(x_1^2) + x_1 x_2(x_2^2)$$

and $x_1 \in \mathbb{K}[x_1, x_2]$, $x_1 x_2 \in \mathbb{K}[x_1, x_2]$.

Quotienting by Ideals

Algebraic Lingo:

"quotienting" = "modding out" = "going modulo"

Let R be a ring and I be an ideal. Then, the quotient ring

$$R/I = \{r + I \mid r \in R\}$$

is basically setting everything in I equal to 0.

Example

$R = \mathbb{K}[x, y, z]$ and $I = (x - y, y - z)$. Then, in $T = R/I$,

$$x - y = 0 \implies x = y \quad \text{and} \quad y - z = 0 \implies y = z$$

$T \cong \mathbb{K}[x]$. Thus R/I has fewer variables to consider than R .

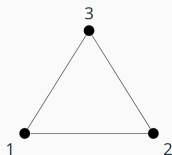
Binomial Edge Ideals

Given a graph $G = (V, E)$ with vertices $V = \{1, 2, \dots, n\}$ and edge set E , the **binomial edge ideal** J_G is defined in the polynomial ring $\mathbb{K}[x_1, \dots, x_n, y_1, \dots, y_n]$ as:

$$J_G = (x_i y_j - x_j y_i \mid (i, j) \in E).$$

Normally, we will describe ideals by their generators.

Example



For K_3 , $E = \{(1, 2), (1, 3), (2, 3)\}$, thus the binomial edge ideal is

$$J_G = (x_1 y_2 - x_2 y_1, x_1 y_3 - x_3 y_1, x_2 y_3 - x_3 y_2)$$

The complete graph K_3 .

The Lexicographic Ordering

For the case of binomial edge ideals, we have the **lexicographic ordering** on monomials, with

$$x_1 > x_2 > \cdots > x_n > y_1 > y_2 > \cdots > y_n$$

We say

$$x_1^{a_1} \cdots x_n^{a_n} y_1^{a_{n+1}} \cdots y_n^{a_{2n}} <_{\text{lex}} x_1^{b_1} \cdots x_n^{b_n} y_1^{b_{n+1}} \cdots y_n^{b_{2n}}$$

if the leftmost entry of $(a_1 - b_1, \cdots, a_{2n} - b_{2n})$ is negative.

Example

$$x_1^2 x_2 y_1 y_2 <_{\text{lex}} x_1^2 x_2^2 y_1^2$$

since

$$(a_1 - b_1, \cdots, a_4 - b_4) = (0, -1, -1, 1).$$

Initial Ideals

The **leading monomial** of a polynomial f with respect to $<$ is the largest monomial in f is denoted $LM_{<}(f)$.

The **initial ideal** of an ideal I with respect to $<$ is the ideal generated by the leading monomials of all of the polynomials in I . Symbolically,

$$\text{in}_{<}(I) = (LM_{<}(f) \mid f \in I).$$

Example

For K_3 ,

$$J_G = (x_1y_2 - x_2y_1, x_1y_3 - x_3y_1, x_2y_3 - x_3y_2)$$

$$\text{in}_{\text{lex}}(J_G) = (x_1y_2, x_1y_3, x_2y_3)$$

Lefschetz Properties

Let $A = A_0 \oplus A_1 \oplus \cdots \oplus A_c$ be a graded finite dimensional $\mathbb{K}$ -algebra, and let $L \in A_1$.

A has the **weak Lefschetz property** (WLP) if there exists L such that for all $i \in [0, c-1]$

$$\times L : A_i \rightarrow A_{i+1}, x \mapsto Lx \quad \text{is injective or surjective.}$$

A has the **strong Lefschetz property** (SLP) if there exists L such that for all $d \in [1, c]$ and $i \in [0, c-d]$

$$\times L^d : A_i \rightarrow A_{i+d}, x \mapsto L^d x \quad \text{is injective or surjective}$$

A graded ring R has SLP if there exists an *l.s.o.p.* Θ so that $A = R/(\Theta)$ has SLP.

Project Findings

Our Main Question

Do polynomial rings modulo the binomial edge ideals have the weak or strong Lefschetz properties?

Binomial edge ideals of what graphs?

- ① Path
- ② Cycle
- ③ Circulant
- ④ Trampoline
- ⑤ Cohen-Macaulay

Path Graphs and Our Magic L.S.O.P.

A **linear system of parameters** (*l.s.o.p.*) $\Theta = (\theta_1, \dots, \theta_d)$ of a graded ring R

- 1 All θ_i have degree equal to 1.
- 2 The ring $R/(\Theta)$ is a finite dimensional vector space.
- 3 d is minimal such that (2) holds.

$$\Theta_{\text{magic}} = \bigcup_{i=1}^{n-1} \{x_i - y_{i+1}\} \cup \{x_n\} \cup \{y_1\}$$

Over a path graph, Θ_{magic} is an *l.s.o.p.*



Figure 2: Path Graph P_5

Theorem (Polymath Jr. 2026)

$\mathbb{K}[x_1, \dots, x_n, y_1, \dots, y_n]/J_{P_n}$ has SLP.

For the path graph P_n ,

$$\frac{\mathbb{K}[x_1, \dots, x_n, y_1, \dots, y_n]}{\text{in}(J_{P_n}) + (\Theta)} \cong \frac{\mathbb{K}[x_1, \dots, x_{n-1}]}{(x_1^2, \dots, x_{n-1}^2)}.$$

Theorem (Stanley's Theorem for $d_i = 2$)

If $\mathbb{K}$ is a field with $\text{char}(\mathbb{K}) = 0$ then $\frac{\mathbb{K}[x_1, \dots, x_n]}{(x_1^2, \dots, x_n^2)}$ has SLP.

This tells us $\mathbb{K}[x_1, \dots, x_n, y_1, \dots, y_n]/\text{in}(J_{P_n})$ has SLP, but by the following theorem

Theorem (Wiebe and Murai)

If S is a polynomial ring and I an ideal of S , then

$$S/\text{in}(I) \text{ has SLP} \implies S/I \text{ has SLP.}$$

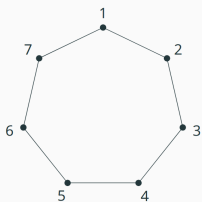
Cycles and circulants

Using the same Θ_{magic} as for path graphs,

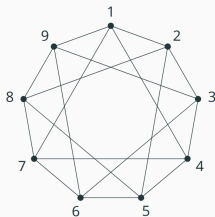
Theorem (Polymath Jr. 2026)

$\mathbb{K}[x_1, \dots, x_n, y_1, \dots, y_n]/J_{C_n}$ has SLP.

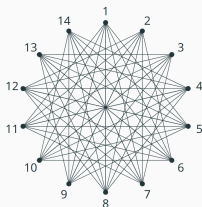
A circulant graph on n vertices with jump set $\{d_1, d_2, \dots, d_s\}$ is the graph where vertices i and j are connected if and only if $j \equiv i \pm d_k \pmod{n}$ for some k .



C_7



$C_9(\{1, 3\})$



$C_{14}(\{4, 5, 6, 7\})$

Failure of WLP

It's not all sunshine and rainbows for the Lefschetz properties of binomial edge ideals.

Let G be the Bowtie graph below. We showed that this graph fails WLP.

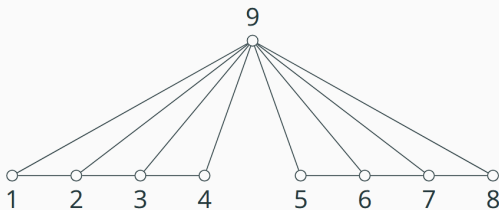


Figure 3: The Bowtie

SLP for Specific Graphs

Additionally, SLP holds for $\mathbb{K}[x_1, \dots, x_n, y_1, \dots, y_n]/J_G$ if:

- 1 G is complete
- 2 G is a path
- 3 G is a complete graph with one or two glued paths.

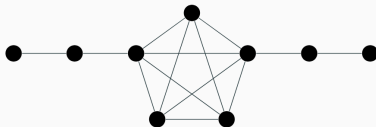


Figure 4: Complete Graph K_5 with two glued paths

WLP for Specific Graphs

Similar to SLP, WLP holds for $\mathbb{K}[x_1, \dots, x_n, y_1, \dots, y_n]/J_G$ if G is two complete graphs joined by a path, with optional paths glued to one or both of the complete graphs, such that the total length of the paths is odd.

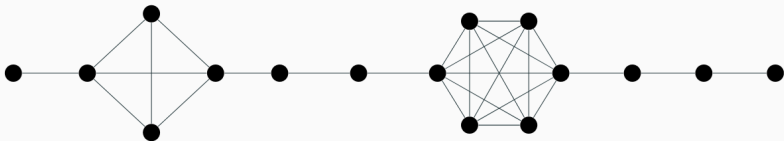
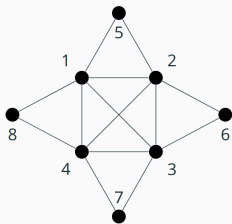


Figure 5: Complete Graphs K_4 and K_6 joined by paths

Trampoline Graphs



Left: the trampoline graph T_4 .
We denote the vertices on the inner complete graph as C and the vertices on the outer ring as R .

A path $i = i_0, i_1, \dots, i_r = j$ from vertices i to j with $i < j$ is called an **admissible path** if, on top of being a path,

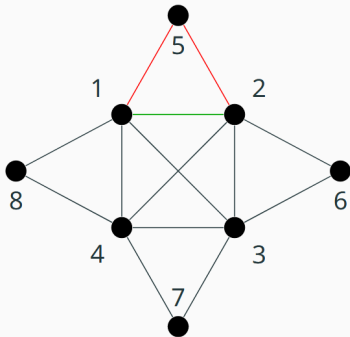
- 1 for each $k \in [1, r - 1]$, either $i_k < i$ or $i_k > j$.
- 2 for any subset $\{j_1, \dots, j_s\} \subsetneq \{i_1, \dots, i_{r-1}\}$, $\{i, j_1, \dots, j_s, j\}$ is not a path.

Trampoline Admissible Paths

Theorem (Polymath Jr. 2026)

The trampoline graph T_n with the earlier labeling has 6 types of admissible paths: CC , CR , CCR , RCC , RCR , and $RCCR$.

For any labeling of T_n , it suffices to check the 6 previous paths in addition to RC . The length of admissible paths in T_n is limited to 4 due to "forbidden" paths CCC , RR , and CRC .



Forbidden path: CRC

The End!

Questions? Comments?